UPC Instruments Pvt. Ltd.

An ISO 9001:2015, 14001-2015 CE & TÜV SÜD Certified Company

Online Water Analyzer

Model No. UPC-WA-202

User Manual

Contents

Introduction

Water Analyzer

Continuous Effluent Monitoring System (Water Analyser) is a device that measures and records the quality of industrial wastewater, emissions from effluent streams. The system uses specialized sensors and instruments to continuously monitor various parameters such as pH, BOD COD, TSS, flow rate and other chemical concentrations required and provides real-time data to ensure that effluent discharge is in compliance with environmental regulations. The data collected by CEMS is often used to optimize treatment processes and improve the sustainability of industrial operations. It works by measuring the absorption or transmission of UV light (200nm to 750nm) by a sample and determining the specific wavelengths of light that are absorbed or transmitted. This information can then be used to identify the presence and concentration of specific chemical compounds in the sample and to study the molecular structure and bonding of the compounds.
The online water analyzer provides reliable and accurate results with low cost of ownership. It is easy to install, with minimal maintenance requirements, and is designed to operate in harsh industrial environments. The analyzer is suitable for a wide range of applications, including wastewater treatment, drinking water, process control, and environmental monitoring.

Important Application

•ETP: to monitor water quality from ETP outlet

•Environmental monitoring: To monitor water quality in rivers, lakes, and oceans.

•Drinking water treatment: To ensure the safety and quality of drinking water.

•Industrial process control: To monitor water quality in industrial processes such as power generation, mining, and food and beverage production.

•Aquaculture: To monitor water quality in fish farms.

•Pool and spa management: To monitor pH and chlorine levels in swimming pools and spas.

•Agriculture: To monitor water quality in irrigation systems and to test soil moisture content.

•Laboratories: To support research and analysis in a variety of fields, including chemistry, biology, and environmental science.

•Plant treatment, Water treatment, Pharmaceutical applications, Biological applications, oxidizing applications etc etc.

IMPORTANT FEATURE

•Real time Monitoring: The system continuously monitor the water quality parameters in real- time and provide updated readings

•Multiple Parameter Analysis: The system measure various water quality parameters such as pH, TSS, BOD, COD, Conductivity, Turbidity, Chlorine, Ammonia, Nitrite, Nitrate, etc.

•Data Logging: The system capability to store and log data for later analysis and reporting.

•Alarm & Notification System: The system generates alarms in case of any critical changes in the water quality parameters.

•User-friendly Interface: The interface is easy to use and understand for operators with limited technical knowledge.

•Remote Access: The systems allow for remote monitoring and control, allowing for real-time monitoring and control from a remote location.

•Scalability: The system is scalable and expandable to accommodate future needs and requirements.

•Reliability: The system is designed for long-term operation and be able to withstand harsh environmental conditions.

•Sensors: The system uses high-quality and accurate sensors to measure the water quality parameters. The sensors should be durable and able to withstand the harsh conditions of an ETP

Data Transmission: The system uses a reliable and secure data transmission method, such as a wired or wireless connection, to transmit the data from the analyzer to the control room.

Data Processing: Robust data processing systems that can handle large amounts of data, perform calculations, and provide real-time results.

Calibration: An automatic calibration feature to ensure accurate measurements. It also allow for manual calibration if needed.

Power Supply: The system should have a reliable and stable power supply to ensure continuous operation.

Environmental Protection: Appropriate environmental protection features to ensure safe and reliable operation in harsh conditions, such as waterproofing, dust proofing, and protection against extreme temperatures.

Interoperability: The system is compatible with other equipment and systems used in the ETP, such as control systems, SCADA systems, and PLCs, to allow for seamless integration. (optional)

Technical Datasheet

Analyser Type	Durable water resistance cabinet
Measuring Principle	COD/ BOD/ TSS: UV vis absorption (Scan between 200nm and 750nm) with xenon lamp source pH: Electrode
Measuring Range	COD: 0 – 500 mg/l, BOD: 0 – 500 mg/l, TSS: 0 – 500 mg/l and pH: 0 – 14 pH
Accuracy	±4% with certified standard
Operating Temperature	0-70 °C
Operating Humidity	4 - 96%
Response Time	≤120 Seconds
Storage	8 GB internal (user can directly copy there data with storage card in excel format)
Response Time	≤60 Seconds
Dispaly	In build human interface touch display with a size of 7 inches
Communication	RS485 or4-20mA
Cloud data	Data is store on cloud space so as user can copy their previous report simply with one click. Unique login credentials for every instrument Data transfer directly to CPCB portal with our channel servers with a help of RTU.
Calibration	Two point calibration is available for calibration
Other specification	Easy user interface, friendly logging and sampling time, trouble shooting

Sensors Specification

Biochemical Oxygen Demand (BOD)
Range: 0 to 500 mg/L
Accuracy: ±2% or ±0.2 mg/L, whichever is greater
Resolution: 0.01 mg/L
Repeatability: ±0.5 mg/L or ±2% of reading, whichever is greater
Reproducibility: ±1 mg/L or ±2% of reading, whichever is greater

Chemical Oxygen Demand (COD)
Range: 0 to 500 mg/L
Accuracy: ±2% or ±10 mg/L, whichever is greater
Resolution: 1 mg/L
Repeatability: ±5 mg/L or ±2% of reading, whichever is greater
Reproducibility: ±10 mg/L or ±2% of reading, whichever is greater

Total Suspended Solids (TSS)
specifications:
Range: 0 to 500 mg/L
Accuracy: ±2% or ±10 mg/L, whichever is greater
Resolution: 1 mg/L
Repeatability: ±5 mg/L or ±2% of reading, whichever is greater
Reproducibility: ±10 mg/L or ±2% of reading, whichever is greater

pH
Method of Analysis: Electrode method
Range: 0 to 14 pH
Accuracy: ±0.1 pH
Resolution: 0.01 pH
Repeatability: ±0.05 pH or ±2% of reading, whichever is greater
Reproducibility: ±0.1 pH or ±2% of reading, whichever is greater

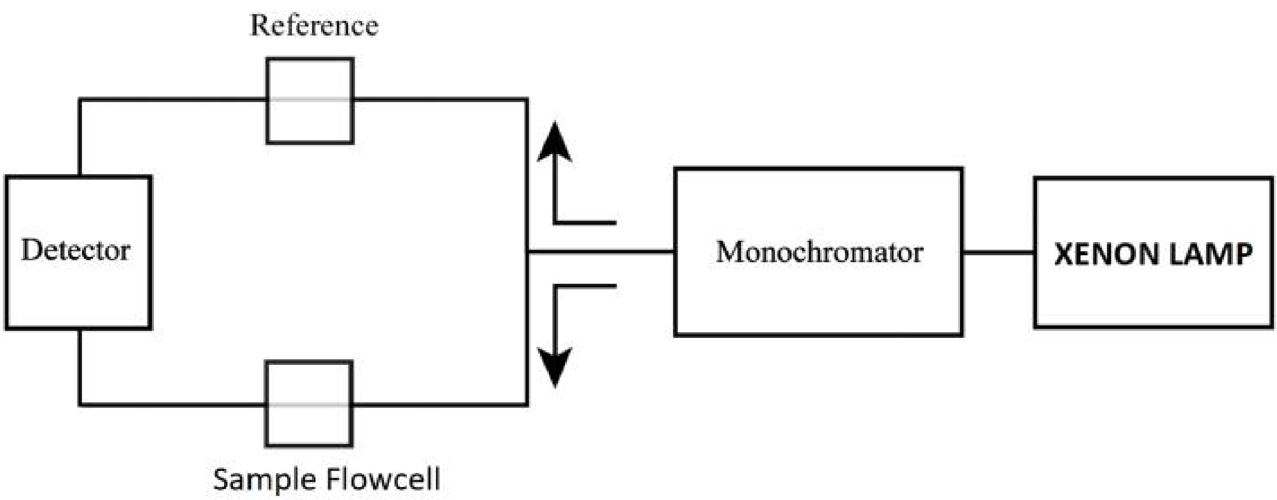
Flow Measurements
Method of Analysis: Electro-Magnetic Flow meter
Range: depend on line size
Accuracy: ±2% or ±0.2 L/min, whichever is greater
Resolution: 0.01 L/min
Repeatability: ±0.1 L/min or ±2% of reading, whichever is greater
Reproducibility: ±0.2 L/min or ±2% of reading, whichever is greater

Working Principal

BOD/COD/TSS/TOC

Our Analyzer is based on UV-Vis absorption spectroscopy. UV-vis absorption spectroscopy is a widely used technique in water analysis due to its high sensitivity and ability to detect a wide range of compounds, including BOD, COD, TSS, TOC and contaminants. UPC online water analyzer provides real-time information.

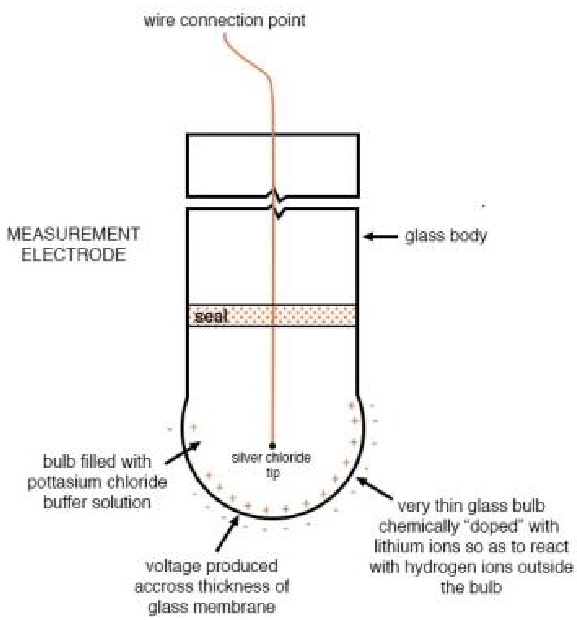
The measurement of parameter in water is based on the absorption of ultraviolet (UV) light source “Xenon lamp” in the water sample. The wavelength of the UV light used for measurement ranges from 200 nm to 750 nm. The measured values are determined from the spectral data. The calculation is based on methods and characteristics that were achieved from a multitude of measurement and longtime analysis and the observation from UV spectra. A Specific design for anti clogging flow cell helps us to calculate accurate data analysis.



pH Measurement

In UPC water analyser pH probe measures the pH (acidity or alkalinity) of a solution by determining the concentration of hydrogen ions (H+) in the solution. The working principle of a pH probe is based on the fact that the hydrogen ion concentration in a solution is directly proportional to its pH.

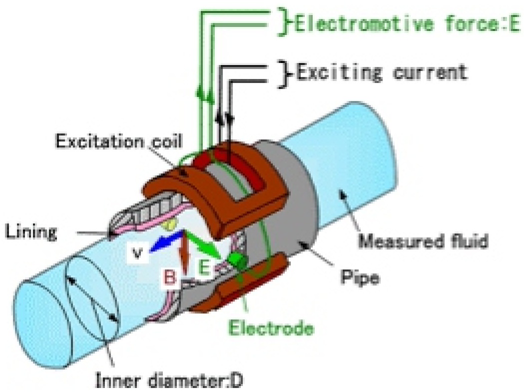
A pH probe consists of a glass electrode and a reference electrode, both of which are placed in the solution to be measured. The glass electrode is sensitive to hydrogen ions and generates a voltage that is proportional to the hydrogen ion concentration in the solution. The reference electrode provides a stable reference voltage and helps to compensate for any changes in temperature or other factors that could affect the measurement.

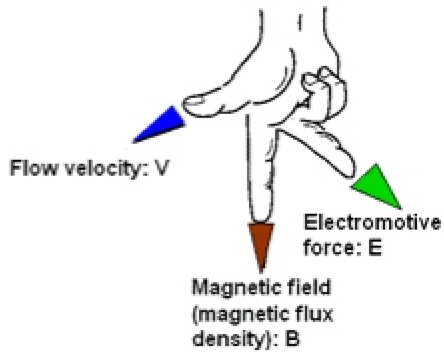


When the pH probe is placed in the solution, the hydrogen ions in the solution diffuse through the glass electrode, generating a voltage that is proportional to the hydrogen ion concentration. This voltage is then measured by a pH meter and used to calculate the pH of the solution.

In summary, the working principle of a pH probe is based on the relationship between the hydrogen ion concentration and the voltage generated by a sensitive glass electrode, which is then used to calculate the pH of a solution

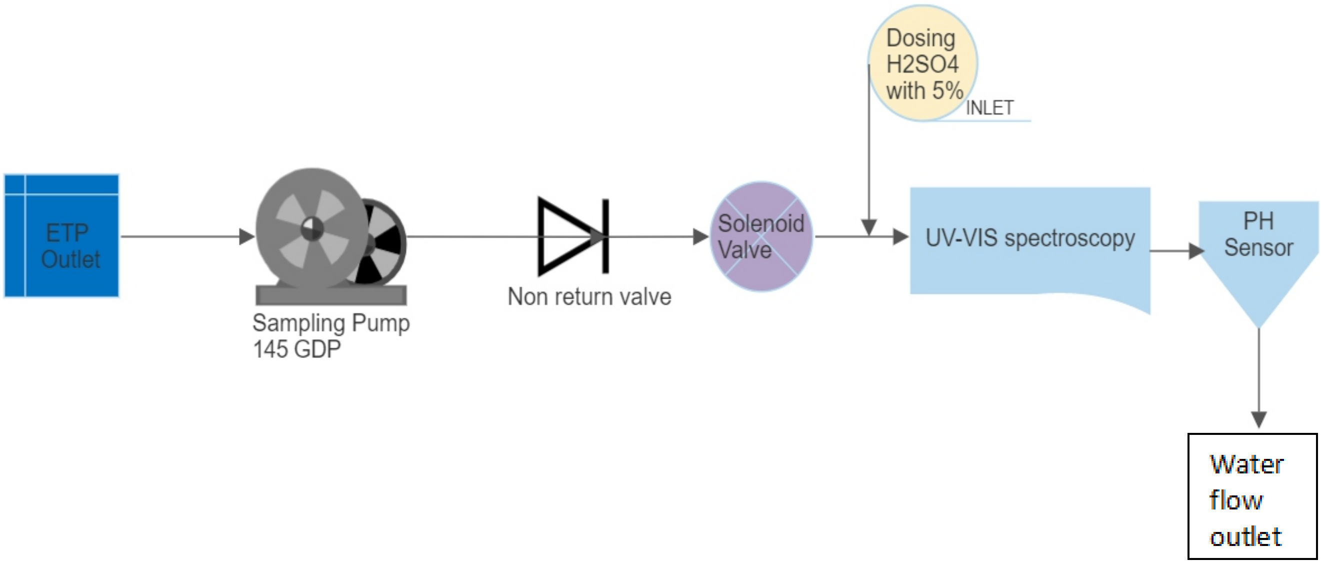
Flow meter



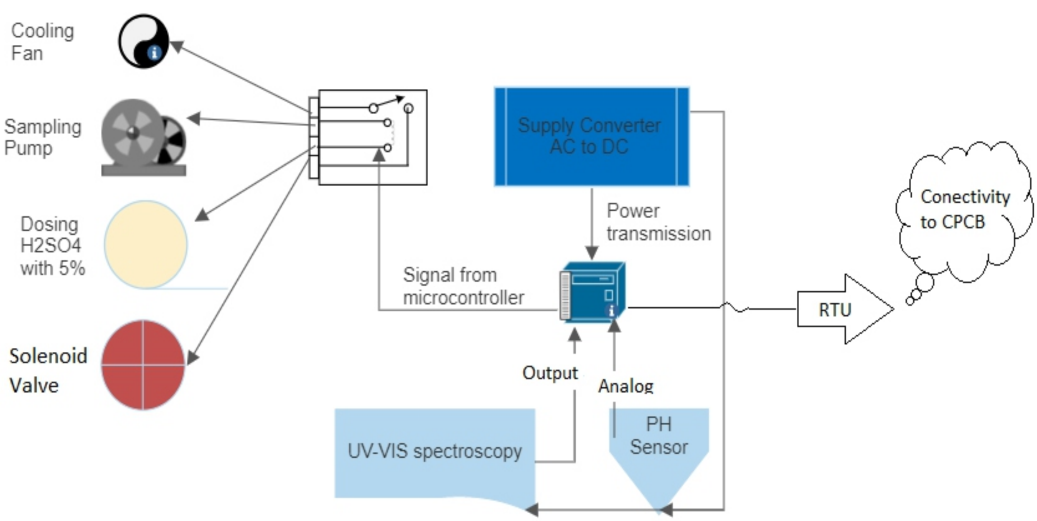


An electromagnetic flow meter (EMF meter) works by measuring the velocity of a conductive fluid in a pipe using electromagnetic technology. It consists of an electrode-lined pipe and an electrical circuit that creates a magnetic field inside the pipe. The fluid flowing through the pipe creates a flow-induced voltage proportional to the flow velocity. This voltage is then measured and converted into a flow rate. The EMF meter is commonly used in industrial processes to measure the flow of liquids in pipelines, such as in water treatment plants, chemical processing, and food and beverage production.

Diagram



Sample flow Diagram



Flow Chart (Electrical Wiring)

Some major Components

•Controller: UPC uses 32bit micro-controller for data Low Power Consumption micro-controller is designed for low power consumption. Large Flash Memory: ESP32 has a large flash memory, ranging from 512 KB to 4 MB, which provides ample storage for applications and data.

•Power supply: we UPC uses 24volt and 5Amps supply with a high efficiency rating to reduce energy waste and having accurate voltage regulation that ensure us stable and consistent output.

•UV-VIS spectroscopy sensor: Gives BOD, COD and TSS output with high accuracy.

•pH sensor : gives us full range of acidic, basic and neutral range.

•Solenoid valve: It works by using an electromagnet to move a plunger that opens or closes the valve. When the electrical current is applied to the solenoid, the plunger is attracted to the electromagnet, opening the valve and allowing the water to flow. When the electrical current is turned off, the plunger returns to its original position, closing the valve and stopping the flow of water.

•Dosing Pump: It is use to clean inner surface where water moves for sampling and we are using 5% concentration for H2SO4 only.

•Sampling Pump: ½ HP pump is used in our water with a 6 meter head and specifically designed for pumping sewage, wastewater, and other liquids containing solids and debris

•Cooling Fan: An essential component that helps to regulate the temperature inside the instrument, thereby preventing overheating and damage to its internal components.

•Relay: It help in switching time for various component like fan, sampling pump, dosing pump and Solenoid valve

•RTU: RTU plays a crucial role in providing real-time environmental data, which is essential for effective pollution control and management. By allowing CPCB officials to monitor environmental conditions in real-time. Its take data from systems output that’s may be RS485 or 4-20 and transmit that data to central pollution control board.

Installations Step

1. **Unpack : open** *packing and match it with attached check list receive in parcel*
2. **Review** *: Now in first step review the equipment specifications, system requirements, and installation site to determine the best location for the instrument, and ensure that the site has the necessary electrical and communication connections. Review all relevant regulations, codes, and standards to ensure that the installation meets all relevant requirements*
3. **Prepare the wall:** *Clean the wall surface and make sure that it is flat and sturdy enough to support the weight of the instrument. If necessary, reinforce the wall by adding additional support, such as brackets or wall anchors.*
4. ***Attach the instrument****: Carefully align the mounting holes of the instrument with the mounting hardware, and attach the instrument to the wall using the screws or other fasteners provided. Tighten the screws securely, making sure that the instrument is level and properly aligned.*
5. **Fix on wall:** *Fix System enclose on wall with suitable location on the wall for the instrument, taking into consideration the access for wiring and maintenance, the visibility of the display, and the environmental conditions.*
6. **RTU connection:** *Connect RTU with system that have RS485 output, it is simply have 2 core wire red or black in where red is "A" & black is "B". Follow marking on RTU and water analyzer system, also gives power to RTU with adaptor*
7. **Sampling Pump attachment:** *Connect wire of Sampling pump coming for Analyzer*
8. **Pipe connection:** *first connect inlet port and sampling pump together with required pipe as per attachment. Secondly attach a pipe to outlet port and drown it ETP*
9. **Suction Pump Installation :***Now It's time to Installed sampling pump in ETP tank, You can simply lowered in bottom of tank after doing electrical and piping connection*
10. *Flow Meter connection: If you have Flow meter to than connect it to 4-20 output coming out from analyzer. If you didn't have ignore this point. Follow steps mention in "point no 13 ii"*

Now it's a time to power up our UPC Water analyzer

UI Interface

11. Wait for a while our analyser start, Now you can see values on touch screen attached with analyser

12. Home Screen is Like UI:1 this in which you can see 5 valve on display BOD, COD, TSS, pH and flow. There is 4 tabs as you see on the top of display

I. Setting

II. Real time graph representation

III. Set clock and Date

IV. calibration

UI:1

13. As we click on first Icon i.e setting than you can see like UI:2 this in which you have 4 option

I. Product info : to know you information like user ID, Serial no, Server ID etc user can easily get then by one click.

II. Flow Calibration: This option is use to calibrate flow meter few simple steps just click to " flow calibration". Than analyser need two values one is zero calibration and second is span calibration. Firstly close pump power observe when flow goes to zero press zero button and secondly start your pump and wait for stability when value become stable set than value as span and press enter.

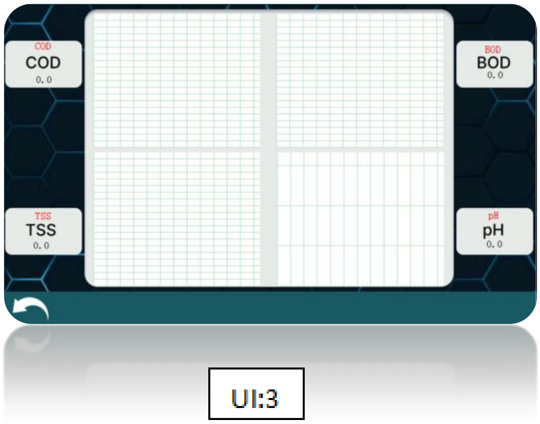
III. Congrats your calibration of flow meter is done

IV. Samp. Time : This option is to change sampling time of sampling pump and you can also do ON/OFF manually with this option as shown in pic.

V. Cleaning time : This option is for cleaning time of doising pump and you can also do ON/OFF manually with this option as shown in pic.

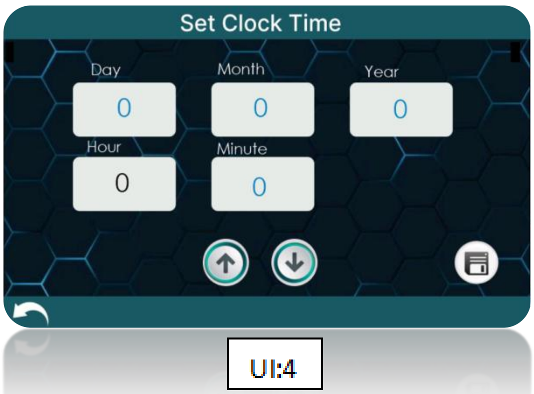
UI:2

14. As you click on second icon user can see UI:3 graph representation as real time values received by analyser



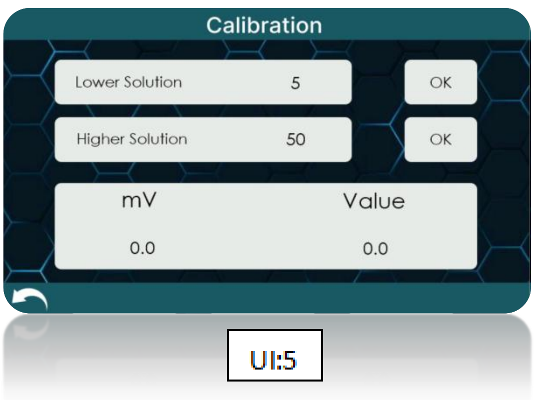
UI:3

15. As you click on third option you can see UI:4 for date and time, where user changes there date and time if it is in correct.



UI:4

16. And with the help of last icon user can calibrates their analyser values UI:5.



UI:5

As per above UI user need 2 sample with different values. Pass lower value sample, feet that pass sample value and press “OK” and then pass higher value, feed its value than press “OK”

Using Cloud data Facility

17. Every Analyser having its unique “USER ID” & “PASSWORD” you can simply login to online.upc.com feed product credential and enjoys cloud facility.
18. Some time user need to integrate other make RTU for online. When its need so user can communicate with the help of Rs485 (datasheet will be provided with our helpline no’s or websites)

For assistant feel free to team *UPC*

WARRANTY

Water Analyzer is warranted against any manufacturing defect for a period of 12 months from date of supply, provided the analyzer has not been misused, damaged, installed for services it is not recommended or the seal has been tempered with. The company shall be liable to furnish part/ parts thereof or full water analyzer as the company may deem fit.

UPC INSTRUMENTS PVT.LTD.
AN ISO 9001:2015, 14001-2015 CE & TÜV SÜD CERTIFIED COMPANY

Office: Plot No. 18, Towel Market, Gohana Road, Panipat-132103 (HARYANA)
Manufacturing Unit: Plot no.125,Near Tehsil, Dinger Majra Road, Gharaunda, Karnal-132114 (HARYANA)
: info@unitechmeter.com : ho@unitechmeter.com : www.unitechmeter.com